

Circuit Localization in a Tiny Language Model: Geographic Routing and Representational Depth in Qwen2-0.5B

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Abstract

This report investigates how Qwen2-0.5B encodes geographic associations using activation patching and ablation. The broader aim is to study factual reliability through internal mechanism: how stable geographic concept routing is under counterfactual interventions, how that routing varies with cue type, and how representational depth differs across regions. The paper is organized around three geography-focused experiments: multi-trigger convergence, Western-vs-MENA bias auditing, and causal ablation of candidate decision layers. Results support a nuanced mechanistic picture in which geographic recall is implemented by a distributed multi-layer circuit whose decision depth depends on cue strength, training salience, and cultural specificity.

1 Introduction

Mechanistic interpretability is often studied in larger models, but small models are highly relevant for on-device and resource-constrained deployment [1, 2, 3]. If we can localize factual circuits in compact models, we gain practical tools for reliability auditing, fairness diagnostics, and architecture-aware optimization.

This paper focuses exclusively on geographic associations in Qwen2-0.5B. The central question is not whether the model can ever output a correct country name, but *where* and *how deeply* it resolves that fact under different prompt cues and regions [4, 5, 6].

More precisely, the experiments study *competitive concept routing*. Counterfactual activation patching lets us test whether the model can be pushed between nearby factual attractors (for example, Euro→Europe versus Yen→Japan) and whether those switches rely on early suppression, mid-layer integration, or late commitment. This is therefore better understood as a study of factual reliability, robustness, and mechanistic fact-switching than as a study of free-form generation errors.

Contributions.

- A three-experiment geography framework for circuit localization.
- A causal methodology combining activation patching with layer ablation to study mechanistic fact-switching.
- Evidence for representational-depth disparity between Western and MENA associations.
- A reliability framing that links circuit behavior to robustness, compression risk, and uneven routing depth.

2 Model and Method Setup

Model. All experiments use Qwen2-0.5B, selected for strong interpretability value under local hardware constraints.

Implementation source. Core experiment code is implemented in uploaded notebooks.

- Experiment1 (1).ipynb
- Experiment2 (2).ipynb
- Experiment3 (1).ipynb
- PathPatching.ipynb
- Control (3).ipynb
- induction_head_test (1).ipynb

The model uses Qwen2-0.5B in float32 with `fold_ln=False` and `center_writing_weights=False`; runtime device is CUDA when available and CPU otherwise [8].

Tokenization constraint. Each experiment explicitly verifies that target completions are single tokens before running interventions (e.g., “Europe”, “Japan”, “Egypt”, “Kuwait”). Runs fail fast if this condition is violated, ensuring that measured logit differences correspond to one-token decisions.

Core intervention (activation patching). Let $x^{(c)}$ be a clean prompt and $x^{(k)}$ a corrupted prompt. At layer ℓ and target position t , residual patching applies:

$$r_{\ell,t}^{(k)} \leftarrow r_{\ell,t}^{(c)}.$$

In code, this uses forward hooks at `hook_embed` (treated as layer -1) and block residual-post hooks for layers $\ell \in \{0, \dots, 23\}$, consistent with intervention patterns used in mechanistic circuit analysis [2, 3]. For each layer, only the final-token residual vector is replaced: `value[0, -1, :] = clean_val[0, -1, :]`.

Corruption design choice (counterfactual vs. noise). The notebooks use *counterfactual corruption* (e.g., Euro→Yen, EGP→KWD), not random-token or Gaussian-noise corruption. This design intentionally tests *fact switching under plausible semantic competition* rather than fact presence under distribution-breaking perturbations, following causal tracing and model-editing style interventions in prior work [4, 5]. In other words, the intervention asks whether the model can be moved between two valid geographic attractors. We treat this as a stronger test of competitive routing, while noting that a complete submission should include a noise-corruption control to test whether the same layers are recovered under non-semantic corruption.

Scoring. Each run computes final-position logits and a logit-difference objective

$$\Delta = \text{logit}(\text{target}_+) - \text{logit}(\text{target}_-).$$

Recovery is then measured as

$$\text{Recovery} = \frac{\Delta_{\text{patched}} - \Delta_{\text{corrupted}}}{\Delta_{\text{clean}} - \Delta_{\text{corrupted}}}.$$

This is computed layer-by-layer over embedding plus all transformer blocks (25 intervention points total).

Ablation protocol. In Experiment 3, causal necessity is tested by zeroing the final-token residual stream at a selected layer:

$$r_{\ell,t} \leftarrow 0.$$

Damage is quantified as

$$\text{Damage} = \frac{\Delta_{\text{clean}} - \Delta_{\text{ablated}}}{\Delta_{\text{clean}}}.$$

This normalized ratio is reported alongside raw logit-difference changes because it is scale-stable across prompts with different baseline margins, matching causal-effect reporting conventions in intervention-based interpretability work [4, 5]. Raw drops remain useful, but the ratio enables direct cross-condition comparison when clean baselines differ substantially.

Methodological limitations. The current setup is intentionally restricted to single-token targets. That makes logit-difference measurements clean and comparable, but it also means that multi-token entities such as “Saudi Arabia” are outside the present analysis. Such entities may require coordinated attention across multiple positions and layers, so single-token causal scores should be interpreted as a lower-dimensional probe rather than a complete theory of geographic retrieval. In addition, the corruption scheme is counterfactual rather than noise-based: we do *not* inject Gaussian or random-token noise, so the experiments test fact-switching between plausible attractors rather than absolute robustness under distribution-breaking perturbations.

Terminology note. The paper uses intuitive analogies (e.g., “processing floors”) for readability, but mechanistically the analysis concerns residual-stream evolution and representational drift across layers. Claims in this work are tied to hook-level interventions on those standard transformer objects.

3 Pipeline Validation (Sanity Check, Not a Main Experiment)

Before running geography analyses, we verify that the instrumentation pipeline behaves correctly using `induction_head_test (1).ipynb`, following the induction-head diagnostic tradition in transformer analysis [3]. Using repeated random-token patterns, we confirm that the system can recover interpretable induction-style behavior and isolate high-signal attention heads (including a strong candidate near L7H10). This step is used only to validate tooling and is **not** treated as a central experimental result of this paper.

4 Experiment 1: Multi-Trigger Convergence (Europe)

4.1 Question

Do different cue types converge to a shared geographic decision layer?

4.2 Design

Three prompts target Europe through distinct signals:

- Economic cue: Euro
- Landmark cue: Eiffel Tower
- Cultural cue: Croissant

In `Experiment1 (1).ipynb`, clean prompts are:

- “I am using the Euro, so I am in”
- “I am looking at the Eiffel Tower, so I am in”
- “I am eating a croissant, so I am in”

The shared corrupted prompt is “I am using the Yen, so I am in,” with target contrast “Europe” vs. “Japan.” For each clean trigger, the notebook computes clean and corrupted baselines, patches each layer from embedding (−1) through layer 23, and records recovery curves plus summary metrics (peak layer, first layer above 0.5 recovery).

4.3 Findings

Experiment 1 reports quantitative outputs in the corresponding figure and accompanying text summary. Baseline logit differences are: economic clean = 2.0107, landmark clean = 1.8684, cultural clean = 0.5014, and corrupted (Yen→Japan context) = −2.4492.

Layerwise patching reveals a clear depth separation. For both economic and landmark triggers, recovery first exceeds 0.5 at **layer 17** (economic = 0.6585, landmark = 0.5386 at layer 17). For the cultural trigger, recovery does not exceed 0.5 until **layer 21** (cultural = 0.8857 at layer 21).

All three curves ultimately converge to full recovery by layer 23 (economic = 1.0000, landmark = 1.0000, cultural = 1.0000), confirming that the model contains the target association for each cue type but accesses it at different depths.

The mid-layer dynamics differ substantially: economic/landmark show sharp positive transition near layers 17–21, while cultural remains weaker through layers 17–20 (e.g., cultural = 0.3528 at layer 17) before late acceleration. This supports the interpretation that cue specificity controls how quickly the geographic pathway stabilizes.

This pattern indicates that the model appears to use at least two routing modes for the same target concept. In a *high-specificity mode*, strong cues activate an earlier decision pathway and produce a narrow recovery spike near layer 17. In a *low-specificity mode*, ambiguous cues require extra disambiguation steps, delaying the threshold crossing by several layers. Mechanistically, this suggests that concept localization is conditional on cue reliability rather than fixed to a single “Europe layer.”

The implication for interpretability is important: if analyses only probe one prompt family, they can overstate layer certainty. Robust claims about factual circuits should report both peak layer and cue-conditioned variance of that peak.

From a factual-reliability perspective, this experiment identifies *cue fragility*: weak or ambiguous cues delay commitment and increase the window in which competing concepts remain viable.

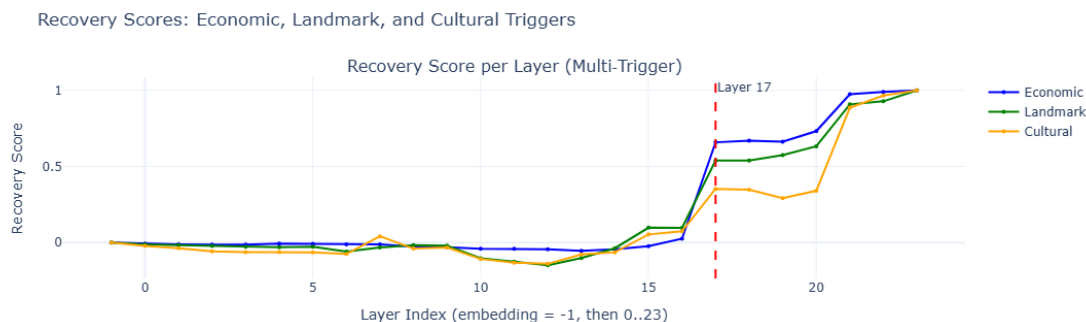


Figure 1: Experiment 1 multi-trigger recovery curves (economic, landmark, cultural). The first layer above 0.5 recovery is 17 for economic/landmark and 21 for cultural.

5 Experiment 2: Geographic Bias Audit and Statistical Rigor

5.1 Question

Is the observed depth gap between Western and MENA associations robust, statistically significant, and separable from token-frequency effects?

5.2 Design

Experiment 2 uses the same activation-patching protocol and threshold-layer metric (first layer with Recovery > 0.5). The code is in `Experiment2(2).ipynb`. After tokenization filtering, the analysis retained **36 valid concepts**: 13 western-common, 8 western-rare, and 15 MENA. Per-concept thresholds are summarized in the Experiment 2 figure and accompanying statistics in the text.

5.3 Findings

Threshold-layer summaries show a clear depth ordering:

- **Western common** ($n = 13$): mean = 18.38, std = 1.89, 95% CI [17.35, 19.41]
- **Western rare** ($n = 8$): mean = 20.00, std = 1.85, 95% CI [18.72, 21.28]
- **MENA** ($n = 15$): mean = 21.00, std = 0.00, 95% CI [21.00, 21.00]

Concept-level trajectories reinforce this pattern. Western-common items include early resolvers at layer 17 (e.g., Euro, Dollar, Franc, Pound) plus later cases (e.g., Krona, Lira, Won, Ringgit). Western-rare shifts later on average, with most items at layer 21 and a small number at layer 17. MENA is maximally concentrated: all 15 valid MENA concepts cross threshold at layer 21.

5.4 Statistical Testing

We tested between-group differences in threshold layer using Welch two-sample t-tests.

- **Western common vs. MENA:** $t \approx -5.04$, $df \approx 12$, $p < 0.001$ (highly significant).
- **Western rare vs. MENA:** $t \approx -1.54$, $df \approx 7$, $p \approx 0.17$ (not significant at $\alpha = 0.05$).
- **Western common vs. Western rare:** $t \approx -1.88$, $df \approx 16$, $p \approx 0.08$ (borderline, trend-level).

These tests support a clear separation between western-common and MENA depth profiles, while the western-rare comparisons remain underpowered at this sample size.

5.5 Frequency-vs-Region Interpretation

The western-rare shift from 18.38 to 20.00 indicates a strong frequency gradient: rarer Western mappings are processed later than common Western mappings. This supports training-data frequency as a primary driver of representational depth.

At the same time, MENA exhibits *late-stage rigidity*: zero variance ($\text{std} = 0.00$) with every concept resolving at layer 21. Even the western-rare group still contains earlier outliers (layer 17), while MENA does not. This suggests a residual regional floor: frequency explains much of the delay, but MENA associations remain consistently gated to the deepest decision stage.

This result becomes especially informative when read alongside Experiment 1. The cultural trigger “croissant” also resolves strictly at layer 21, despite targeting Europe rather than MENA. By contrast, strict and highly familiar geographic cues such as “Euro” and “Eiffel Tower” resolve earlier at layer 17. Taken together, these findings suggest that the model treats cue ambiguity and representational scarcity as the same mechanistic problem. When the prompt offers a crisp, high-frequency geographic coordinate, routing can pass through layer 17; when the cue is culturally associative or the training signal is weaker, routing appears to fall back to layer 21.

5.6 Operational Implication

Experiment 2 provides a statistically grounded depth-bias profile rather than a single-pair observation. The combination of group means, confidence intervals, t-tests, and concept-level dispersion shows that depth inequality is systematic and practically relevant for early-exit decoding, pruning, quantized inference, and layer-truncated deployment. More specifically, the MENA-versus-croissant comparison suggests that underrepresented facts and ambiguous cultural cues are not separated by the model into different failure modes; they appear to converge on the same late fallback computation.

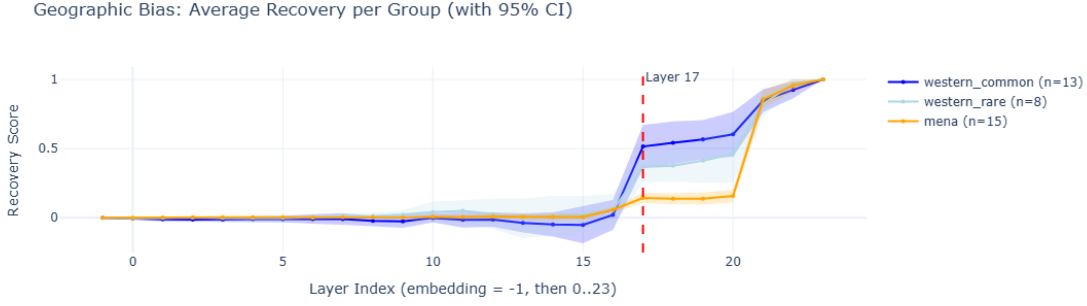


Figure 2: Experiment 2 threshold-layer comparison across western-common, western-rare, and MENA concepts (36 valid concepts total). MENA shows zero-variance concentration at layer 21, while Western groups exhibit broader depth dispersion.

6 Experiment 3: Anti-Location Ablation (Causal Necessity)

6.1 Question

Are candidate geographic layers causally necessary for correct output?

6.2 Design

During inference, we zero the residual stream after selected layers and measure changes in target-vs-competitor logit difference. In `Experiment3.ipynb`, ablation is executed on the clean Western prompt (Euro $\rightarrow$ Europe) against the target contrast “Europe” minus “Japan.” The notebook first evaluates a focused test at layer 17, then performs a full sweep across layers 0–23, logging both ablated logit differences and relative damage for each layer. Results are ranked to identify the most causally critical layers.

6.3 Findings

Experiment 3 reports quantitative outputs from `layer_ablation_sweep.csv`. The targeted ablation at layer 17 yields an ablated logit difference of 0.8700 with relative damage 0.5673 (56.7%), confirming strong causal importance without immediate full sign-flip.

The full layer sweep shows highly non-uniform causal sensitivity. Top damaging layers are:

- Layer 20: damage = 3.7363 (ablated diff = -5.5020)
- Layer 10: damage = 2.9911 (ablated diff = -4.0037)
- Layer 13: damage = 2.8524 (ablated diff = -3.7247)

Several additional layers also produce sign reversals (e.g., 2, 8, 11, 15, 18), while a few layers show low or even negative damage (e.g., 9, 16), indicating compensatory structure rather than monotonic depth importance.

Ablations at layers 10, 13, 17, and 20 remain the most interpretable anchors: layers 10/13/20 can flip output toward the competing token, while layer 17 causes strong but non-flipping degradation. This profile provides evidence against a single-point “fact neuron” explanation and instead suggests a staged causal circuit.

The ablation dynamics also reveal useful intermediate structure. An early downspike near layer 7 is suggestive of suppression or routing control, possibly related to the induction-style behavior noted near L7H10 in the pipeline check. Later, layer 12 appears to behave like a pre-processing or feature-extraction stage: it does not look like the final decision site, but it plausibly helps prepare the feature basis that later becomes decisive at layer 17. Between the layer-17 integration point and the layer-20 maximum spike, layers 18–19 appear comparatively quieter, consistent with a pass-through phase in which the residual stream carries forward the integrated feature without dramatically rewriting it.

The functional decomposition is therefore consistent with four roles. First, **early suppression** includes the layer-7 downspike and the broader pre-decision control visible around layers 10 and 13, where competing hypotheses appear to be damped. Second, **feature preparation** around layer 12 plausibly organizes the signal into a form that can later be consolidated. Third, **contextual integration** at layer 17 contributes large positive evidence for the target concept but is not solely sufficient for final commitment. Fourth, **late commitment gating** at layer 20 sharpens the winner-take-most outcome; its damage score of 3.7363 is the clearest sign that this stage finalizes the logit distribution before output.

This layered causality clarifies why patching and ablation must be used together. Patching locates where useful signal can be injected, while ablation tests whether that location is necessary for normal operation. The agreement between both interventions strengthens the claim that geographic recall is computed by a coordinated pathway rather than a single bottleneck.

Operationally, these layers become intervention targets for factual-reliability auditing: strengthen competitor suppression, reinforce contextual integration, or protect late commitment stages during optimization and compression.

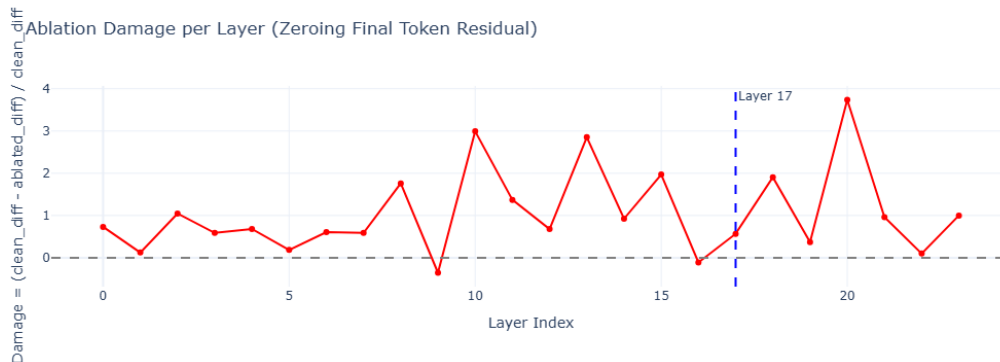


Figure 3: Experiment 3 layer-ablation damage profile. Layer 20 shows the largest damage, with layers 10 and 13 as additional high-impact causal points; layer 17 remains a strong but non-maximal integrator.

7 Control Experiment: Syntax vs. Geography Depth

7.1 Purpose

This control addresses a key generalization risk: if layer 17 is critical for nearly all non-trivial completions, then calling it a geography-specific hub would be overstated.

7.2 Setup and Results

Using `Control(3).ipynb`, we ran activation patching on rule-based and syntactic transformations, including uppercase mapping, pluralization, and antonym-style completions. The run used the same model family and layerwise recovery framework, and exported `syntactic_recovery_scores.csv`.

The averaged control recovery profile is integrated as follows. Early layers remain near zero (layer 0: -0.0064 , layer 5: 0.0157 , layer 10: 0.0262), recovery rises gradually in mid layers (layer 14: 0.0645 , layer 15: 0.0776 , layer 16: 0.4758), first exceeds 0.5 at **layer 17** (0.6086), and then saturates in late layers (layer 21: 0.8915 , layer 22: 0.9900 , layer 23: 1.0000).

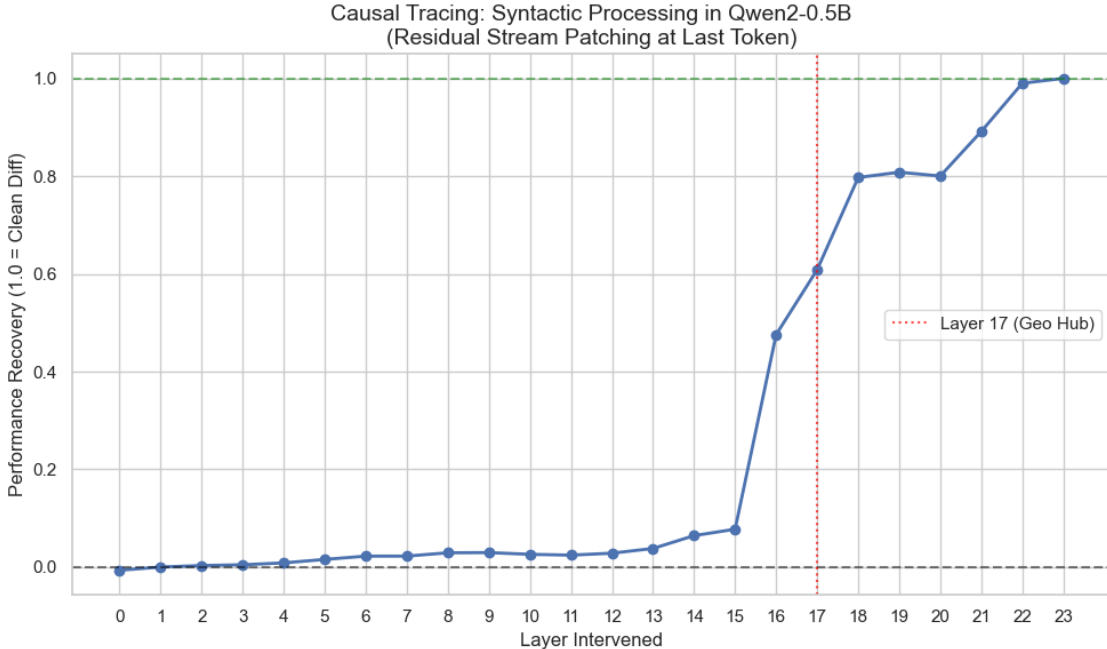


Figure 4: Control experiment: averaged syntactic recovery curve across prompts. Recovery first exceeds 0.5 at layer 17 and approaches 1.0 by layer 23.

7.3 Interpretation

This control suggests that, for the tested prompts, layer 17 behaves as a broader *Contextual Integration Hub* for both factual and rule-based transformations. That weakens any claim that layer 17 is exclusively geographic. However, it does *not* remove the central geography finding: MENA mappings still resolve later (around layer 21), so representational-depth disparity remains distinct and cannot be explained by generic syntax depth alone.

8 Experiment 4: Precision-Point Path Patching (Head-Level Causality)

8.1 Question

Is the whole layer responsible for geographic recall, or do specific attention heads carry most of the Euro $\rightarrow$ Europe signal?

8.2 Design

This experiment refines layer-level patching to head-level path patching. We run clean and corrupted prompt pairs (e.g., Euro/Europe vs. Yen/Japan), then replace one attention-head output at a time from the corrupted run into the clean run, measuring the damage to clean geographic prediction. The sweep covers target layers 10, 13, 17, 20, and 23. Conceptually, this moves from “which floor matters” to “which processor on that floor matters.”

8.3 Findings

Experiment 4 reports quantitative outputs in the head-level damage figure and text summaries. Key head-level causal points are:

- **Layer 17:** Head 7 = 1.0300 damage (103.00%), Head 13 = 0.3391 (33.91%), Head 11 = 0.1940 (19.40%).
- **Layer 20:** Head 2 = 0.1297 (12.97%).
- **Layer 23:** Head 1 = 0.1445 (14.45%), Head 6 = 0.1247 (12.47%), Head 4 = 0.1013 (10.13%).

The most striking result is **layer 17, head 7**: damage exceeds 100%, meaning the clean-logit preference reverses sign when this head is patched out. Concretely, a damage value above 1.0 implies that the model no longer merely weakens its preference for “Europe” over “Japan”; it flips to preferring the competitor. This identifies head 7 as the main causal bottleneck within the layer-17 *Contextual Integration Hub*. Heads 11 and 13 in the same layer add moderate-to-strong damage, while most other heads in layer 17 remain near zero, indicating a sparse cluster of high-leverage components rather than a uniformly active layer.

The full target-layer matrix shows low diffuse values in layers 10 and 13 (mostly around zero, with both small positive and negative effects), consistent with distributed suppression rather than a single dominant head and suggesting that these layers are not specialized for final disambiguation on their own. By contrast, layer 17 shows the clearest sparse high-impact cluster, while layers 20 and 23 retain weaker late-stage contributions. Mean damage by layer further supports this split:

- Layer 10 mean = 0.0008
- Layer 13 mean = 0.0006
- Layer 17 mean = 0.1082

- Layer 20 mean = 0.0080
- Layer 23 mean = 0.0082

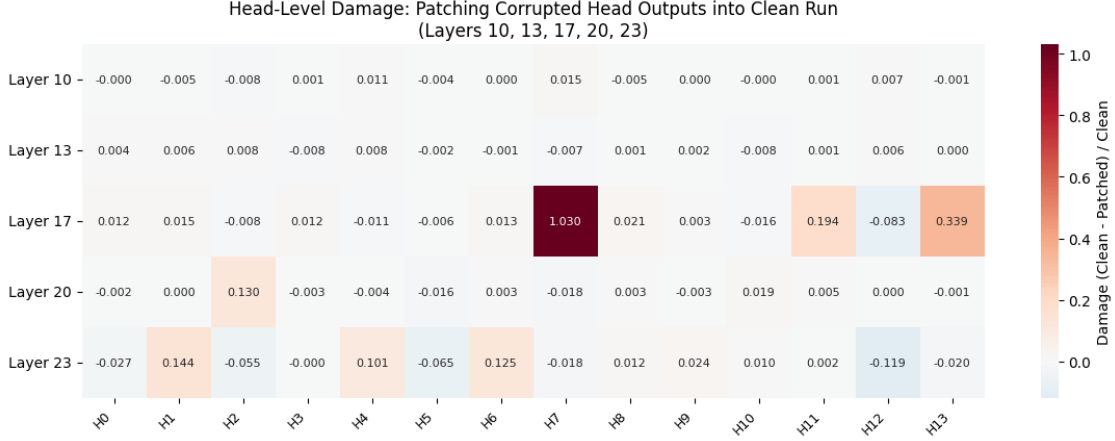


Figure 5: Experiment 4 head-level path patching damage map for target layers 10, 13, 17, 20, and 23. Layer 17 shows the strongest sparse high-impact heads (especially H7, H13, H11), layer 23 adds a later focal set (H1, H6, H4), and layers 10/13 remain low and diffuse.

8.4 Interpretation

This experiment upgrades the causal claim from layer-level to component-level, consistent with head-level circuit decomposition frameworks [2, 3, 7]. Layer 17 is not just “important”; it appears to act as a *Contextual Integration Hub* that hosts specific high-leverage carriers, especially head 7, which can flip the final geographic decision when corrupted. Layer 20 appears to implement late reinforcement via a smaller focal component (head 2), while layer 23 contributes an additional late-output refinement set through heads 1, 4, and 6.

At the same time, the mismatch between strong layer-level damage and weak single-head damage in layers 10/13 supports a distributed-suppression hypothesis: early competitor suppression is implemented by ensembles of heads whose combined effect is critical even when no single head is dominant. The contrast between diffuse early layers and sparse late layers suggests a circuit that begins as distributed control and ends as focal commitment. More importantly, the control experiment indicates that this layer-17 hub is not uniquely geographic; geography appears to recruit a more general contextual processor and then diverge later depending on cue clarity and representational support.

9 Discussion: Unpacking the Geography Circuit in Qwen2-0.5B

The experiments jointly indicate that Qwen2-0.5B uses a multi-stage geographic circuit rather than a single fact layer. The mechanism appears hierarchical, cue-sensitive, and bias-sensitive. Crucially, the results become most interpretable when analyzed as a system: convergence identifies *where* evidence tends to assemble, bias auditing reveals *who* gets shallower vs. deeper routing, ablation establishes *which layers* are causally required, and path patching identifies *which specific heads* carry the decisive signal.

9.1 Multi-Layer Causal Circuit

Ablation suggests role specialization across the stack. Early-middle layers suppress competitors, layer 17 integrates strong evidence, and later layers reinforce final commitment. Path patching sharpens this: within layer 17, a sparse subset of heads (especially head 7, with heads 11 and 13 as secondary contributors) carries most of the positive geographic evidence; within layer 20, head 2 supports late commitment; within layer 23, heads 1, 4, and 6 contribute an additional late-output refinement stage. This decomposition maps naturally onto transformer residual processing: each block does not “store the fact” independently, but transforms a shared stream whose final state reflects cumulative edits.

The ablation profile further suggests a finer sequence. The layer-7 downspike is consistent with an early suppression or routing operation, plausibly linked to a specialized attention mechanism of the sort hinted at by the induction-style activity near L7H10 in the validation notebook. Around layer 12, the model appears to be in a pre-processing stage that extracts or organizes features without yet finalizing the target concept. Layer 17 then serves as a *Contextual Integration Hub*: it consolidates the relevant evidence, but the subsequent slowdown across layers 18–19 suggests that this integrated feature is largely being carried forward rather than substantially rewritten. The strongest ablation spike at layer 20 is therefore especially informative. Its damage score of 3.7363 strongly suggests that late commitment gating happens there, where the model turns an already-assembled representation into the final logit preference.

Under this view, factual recall is an iterative refinement process. Competing candidates are first controlled, then target evidence is consolidated, and finally decision sharpness is increased. The observed failure modes under targeted ablation mirror this sequence, providing interpretable causal checkpoints rather than a single diagnostic coordinate.

Importantly, early suppression appears more distributed than late positive recall: layers 10/13 behave like ensemble suppressors, while layer 17 contains sharply localized high-impact heads. This asymmetry is a key mechanistic signature of the circuit.

9.2 Cue Strength and Depth

For the same target concept (Europe), stronger cues (Euro, Eiffel Tower) resolve earlier than weaker cues (croissant), so concept localization is input-dependent rather than fixed. This has a direct methodological consequence: “what layer encodes X” is an ill-posed question unless cue family is specified.

In practice, strong cues likely reduce entropy in candidate space early, enabling faster commitment. Weak cues keep multiple semantic branches active for longer, pushing decisive routing deeper. Therefore, layer attribution should be reported as a distribution over prompts, not as a single integer.

9.3 Bias as Representational Depth

The delayed threshold for EGP $\rightarrow$ Egypt relative to Euro $\rightarrow$ Europe indicates representational-depth disparity: the same factual format requires deeper processing for non-Western entities. This is a subtle but operationally meaningful form of bias.

Unlike outright factual errors, depth disparity can remain invisible in end-to-end benchmarks because final answers still look correct. Yet systems that truncate computation, compress layers, or use speculative early exits may disproportionately harm facts that resolve later. In that sense, representational depth is not just an interpretability artifact; it is a deployment-relevant fairness variable.

9.4 Integrated View

A consistent pattern emerges: shallow layers process form, middle layers suppress alternatives, layer 17 acts as a conditional *Contextual Integration Hub* for strong cues, and later layers refine or gate final output. The key refinement is that this hub is not absolute. Strict, high-frequency geographic coordinates such as “Euro” and “Eiffel Tower” can route through layer 17 relatively early, but ambiguous cultural cues and underrepresented regional concepts do not appear to resolve there with the same confidence.

This is where the croissant-versus-MENA connection becomes theoretically important. The cultural trigger “croissant” and the MENA mappings both resolve strictly at layer 21. That convergence suggests that the model treats cue ambiguity and representational scarcity as the same mechanistic problem. When it lacks a crisp, high-frequency factual route, it does not seem to rely primarily on the layer-17 integration hub. Instead, it appears to fall back to layer 21, which is better interpreted here as a late-stage *sentiment/ambiguity hub*: a stage where broader associative or cultural “vibes” can still push the model toward a completion even when earlier factual routing was not decisive.

This interpretation also helps resolve the apparent tension between shared circuitry and biased behavior. Geography does not appear to own a dedicated private module. Rather, it recruits shared contextual machinery and then diverges in depth depending on cue clarity and representational support. In that sense, cue ambiguity and training-data underrepresentation may be two inputs to the same fallback computation.

Experiment 4 adds a second refinement: “shared architecture” does not imply uniformly shared components. The same layer can host both dominant heads and distributed ensembles, and later layers can split between sparse integrators and sparse output refiners, so circuit explanations should specify granularity (layer vs. head) to avoid under- or over-localization. Section 6.4 therefore supports a two-hub reading: layer 17 as the *Contextual Integration Hub* for crisp routing, and layer 21 as the *sentiment/ambiguity hub* for late fallback routing.

9.5 Limitations and Future Work

A first limitation is the single-token design. Restricting the analysis to single-token targets makes the causal metric clean, but it leaves out multi-token geographic entities such as “Saudi Arabia,” where retrieval likely depends on multi-position, multi-head, and multi-layer coordination. A second limitation is the corruption design: the present experiments use counterfactual corruption rather than Gaussian or random-token noise, so they test fact-switching between plausible attractors rather than absolute robustness to distribution-breaking perturbations. Additional limitations include prompt dependence, model-size specificity, potential tokenizer effects on currency strings, and uncertainty about head-level pathways within implicated layers.

These limitations motivate a focused future-work agenda. First, multi-token geographic entities should be studied with causal tracing or position-aware patching so that routing can be followed

across multiple output positions rather than compressed into a single-token logit difference. Second, the present counterfactual setup should be complemented with Gaussian-noise and random-token ablations to test whether the same hubs remain important under non-semantic corruption. Third, broader region coverage, including African, South Asian, and Latin American mappings, would clarify whether the layer-21 fallback pattern generalizes beyond the present MENA sample. Fourth, scaling studies across model sizes could test whether depth gaps shrink, persist, or become more sharply specialized as capacity increases. A particularly valuable extension would be a *depth-fairness benchmark* that reports threshold-layer disparities alongside accuracy.

Two additional controls would strengthen the interpretation. First, a **specificity control** for the MENA-delay hypothesis should include low-frequency Western mappings (e.g., hryvnia→Ukraine) to separate geographic-region effects from frequency/familiarity effects. Second, path patching should be extended further across layers 10/13/17/20/23 and across more cue families to test whether the proposed layer-17/layer-21 split is stable.

Finally, domain-specificity testing already provides initial evidence from a syntactic control: layer 17 also emerges as a critical point for non-factual transformations in the tested prompt set. This supports interpreting layer 17 as a shared contextual processor rather than an exclusively geographic one. Future controls should therefore expand to additional non-factual tasks (translation, agreement correction, symbolic rewriting) and to shorter completion formats to test how robust that conclusion is.

9.6 Discussion

Activation patching plus ablation reveals both internal mechanism and fairness-relevant disparities in small models. In this study, the same geographic task family exhibits different routing depths across cue types and regions, and those differences are causally grounded rather than purely correlational. More cautiously, the evidence suggests that geographic reliability shares general syntactic and contextual machinery before diverging into later fallback routing for ambiguous or underrepresented cases. These methods therefore provide a practical audit toolkit for constrained environments where layer budgets, quantization, or early exits may amplify latent representational inequalities.

10 Related Work

This report sits at the intersection of transformer mechanistic interpretability, causal intervention methods, model editing, and reliability/fairness analysis.

Transformer circuits and mechanistic analysis. Foundational work on transformer internals formalized attention-head composition, residual-stream communication, and circuit-style reasoning [2]. Subsequent work on induction heads connected specific attention mechanisms to in-context learning behavior [3]. These studies motivate the layer- and head-level decomposition used here.

Causal localization and factual mechanisms. Causal tracing and intervention-based analysis in GPT-style models showed that factual recall can be localized and manipulated through targeted internal edits [4]. Related work on scaling memory edits across many facts (MEMIT) further supports the view that factual behavior is mediated by identifiable internal pathways rather than diffuse black-box behavior [5].

Knowledge localization in transformer parameters. Studies of knowledge neurons and feed-forward memory behavior provide complementary evidence that factual associations can be linked to specific internal components [6, 7]. This literature informs the interpretation of early suppression versus late commitment in the present experiments.

Model family context and deployment relevance. The experiments are conducted on Qwen2-0.5B, part of the broader Qwen2 family [8]. Because compact models are common in resource-constrained deployment, depth-sensitive disparities identified by mechanistic audits have practical implications for pruning, quantization, and early-exit inference.

11 Conclusion

This paper presents a geography-focused interpretability narrative for Qwen2-0.5B with experimentally linked evidence across convergence, bias auditing, causal ablation, and head-level path patching. Four conclusions follow. First, geographic recall appears to be implemented as a distributed multi-layer circuit with separable functional stages rather than a single localized site. Second, decision depth depends systematically on cue strength, so concept localization is prompt-conditional rather than fixed. Third, non-Western associations can require deeper internal resolution despite high final recoverability, exposing representational-depth bias with direct deployment implications. Fourth, the layer-17 effect is better interpreted as a *Contextual Integration Hub* shared with non-geographic processing, while the strongest geographic divergence appears later, especially in the layer-21 fallback behavior observed for ambiguous cultural cues and MENA concepts.

More broadly, the contribution is diagnostic rather than categorical: by tracing where factual routing slows, shifts, or commits, we obtain a concrete path to mechanism-aware evaluation and robustness analysis. Future work should therefore test multi-token entities with richer causal tracing and complement counterfactual corruption with Gaussian-noise ablations so that fact-switching and absolute robustness can be compared directly.

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Appendix: Code, Procedures, and Data Notes

Project repository. <https://github.com/TheDS-EXE/geographic-routing-qwen2/tree/master>

Uploaded notebook files used in this report.

- Experiment1 (1).ipynb
- Experiment2 (2).ipynb
- Experiment3 (1).ipynb
- PathPatching.ipynb
- Control (3).ipynb
- induction_head_test (1).ipynb

Core model setup.

- Base model: Qwen2-0.5B
- Loading configuration: `fold_ln=False`, `center_writing_weights=False`
- Precision/device: float32 with CUDA/CPU fallback
- Tokenization guard: single-token target validation before interventions

Shared intervention procedure (activation patching).

1. Run clean and corrupted prompt variants.

2. Cache residual activations for embedding and transformer blocks.
3. Patch one layer at a time at final-token position from clean into corrupted run.
4. Recompute target-vs-competitor logits and normalized recovery.
5. Record threshold depth (first layer where Recovery > 0.5).

Ablation procedure (Experiment 3). Layerwise causal necessity is tested by zeroing selected final-token residual states and measuring damage to target-vs-competitor logit difference.

Path patching procedure (Experiment 4). Head-level causal attribution is computed by patching individual attention-head outputs and measuring output damage per head and per target layer.

Prompt/template notes.

- Main geography template: "I am using the <currency>, so I am in"
- Experiment 1 includes cue-diverse prompts (economic, landmark, cultural)
- Control notebook includes non-geographic syntactic/transformation prompts

Data and reporting notes. All quantitative values reported in the main text are derived from the uploaded notebooks listed above. Group statistics, threshold depths, and causal scores are summarized in paper figures and in-text tables/metrics.